Unit 7

Operating System 6 hours

7.1 Introduction
7.2 History and evolution of OS
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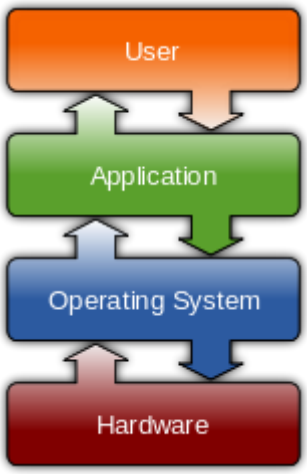
Introduction

An operating system, or "OS," is software that communicates with the hardware and allows other programs to run. An **operating system** is a collection of software that manages computer hardware resources and provides common services for computer programs. **Operating system** is system software that controls the operation of a computer, directs the input and output of data, keeps track of files, and controls the processing of computer programs. Its roles include managing the functioning of the computer hardware, running the applications programs, serving as an interface between the computer and the user, and allocating computer resources to various functions.



Linux, Windows 2000, VMS, OS/400, AIX, and z/OS are all examples ofoperating systems.

Some of the services provided by OS are:

- In a multitasking operating system where multiple program can be running at the same time, the operating system determines which applications should run in what order and how much time should be allowed for each application before giving another application a turn.
- It manages the sharing of internal memory among multiple applications.
- It handles input and output to and from attached hardware devices, such as hard disks, printers, anddial-up ports.

History and evolution of OS / Generation of Operating System

An operating system is a type of software that acts as an interface between the user and the hardware. It is responsible for handling various critical functions of the computer or any other

machine. Various tasks that are handled by OS are file management, task management, garbage management, memory management, process management, disk management, I/O management, peripherals management, etc.

Below are four generations of operating systems.

1. The First Generation
2. The Second Generation
3. The Third Generation
4. The Fourth Generation

1. The First Generation (1940 to early 1950s)

In 1940, an operating system was not included in the creation of the first electrical computer. Early computer users had complete control over the device and wrote programs in pure machine language for every task. During the computer generation, a programmer can merely execute and solve basic mathematical calculations. an operating system is not needed for these computations.

2. The Second Generation (1955 – 1965)

GMOSIS, the first operating system (OS) was developed in the early 1950s. For the IBM Computer, General Motors has created the operating system. Because it gathers all related jobs into groups or batches and then submits them to the operating system using a punch card to finish all of them, the second-generation operating system was built on a single-stream batch processing system.

3. The Third Generation (1965 – 1980)

Because it gathers all similar jobs into groups or batches and then submits them to the second generation operating system using a punch card to finish all jobs in a machine, the second-generation operating system was based on a single stream batch processing system. Control is transferred to the operating system upon each job's completion, whether it be routinely or unexpectedly. The operating system cleans up after each work is finished before reading and starting the subsequent job on a punch card. Large, professionally operated machines known as mainframes were introduced after that. Operating system designers were able to create a new operating system in the late 1960s that was capable of multiprogramming—the simultaneous execution of several tasks in a single computer program. In order to create operating systems that enable a CPU to be active at all times by carrying out multiple jobs on a computer at once, multiprogramming has to be introduced. With the release of the DEC PDP-1 in 1961, the third

generation of minicomputers saw a new phase of growth and development.

4. The Fourth Generation (1980 – Present Day)

The fourth generation of personal computers is the result of these PDPs. The Generation IV (1980–Present) The evolution of the personal computer is linked to the fourth generation of operating systems. Nonetheless, the third-generation minicomputers and the personal computer have many similarities. At that time, minicomputers were only slightly more expensive than personal computers, which were highly expensive.

The development of Microsoft and the Windows operating system was a significant influence in the creation of personal computers. In 1975, Microsoft developed the first Windows operating system. Bill Gates and Paul Allen had the idea to advance personal computers after releasing the Microsoft Windows OS. As a result, the MS-DOS was released in 1981, but users found it extremely challenging to decipher its complex commands. Windows is now the most widely used and well-liked operating system available. Following then, Windows released a number of operating systems, including Windows 95, Windows 98, Windows XP, and Windows 7, the most recent operating system. The majority of Windows users are currently running Windows 10. Apple is another well-known operating system in addition to Windows.

Types of Operating System

Operating Systems have evolved in past years. It went through several changes before getting its original form. These changes in the operating system are known as the evolution of operating systems. OS improve itself with the invention of new technology. Basically, OS added the feature of new technology and making itself more powerful. Let us see the evolution of operating system year-wise in detail:

1. No OS – (0s to 1940s)
2. Batch Processing Systems -(1940s to 1950s)
3. Multiprogramming Systems -(1950s to 1960s)
4. Time-Sharing Systems -(1960s to 1970s)
5. Introduction of GUI -(1970s to 1980s)
6. Networked Systems – (1980s to 1990s)
7. Mobile Operating Systems – (Late 1990s to Early 2000s)
8. AI Integration – (2010s to ongoing)

1. No OS – (0s to 1940s)

As we know that before 1940s, there was no use of OS. Earlier, people are lacking OS in their computer system so they had to manually type instructions for each task in machine language (0-1 based language). And at that time, it was very hard for users to implement even a simple task. And it was very time consuming and also not user-friendly. Because not everyone had that much level of understanding to understand the machine language and it required a deep understanding.

2. Batch Processing Systems -(1940s to 1950s)

With the growth of time, batch processing system came into the market. Now Users had facility to write their programs on punch cards and load it to the computer operator. And then operator make different batches of similar types of jobs and then serve the different batch (group of jobs) one by one to the CPU. CPU first executes jobs of one batch and them jump to the jobs of other batch in a sequence manner.

3. Multiprogramming Systems -(1950s to 1960s)

Multiprogramming was the first operating system where actual revolution began. It provides user facility to load the multiple programs into the memory and provide a specific portion of memory to each program. When one program is waiting for any I/O operations (which take much time) at that time the OS give permission to CPU to switch from previous program to other program (which is first in ready queue) for continuous execution of program with interrupt.

4. Time-Sharing Systems -(1960s to 1970s)

Time-sharing systems is extended version of multiprogramming system. Here one extra feature was added to avoid the use of CPU for long time by any single program and give access of CPU to every program after a certain interval of time. Basically, OS switches from one program to another program after a certain interval of time so that every program can get access of CPU and complete their work.

5. Introduction of GUI -(1970s to 1980s)

With the growth of time, Graphical User Interfaces (GUIs) came. First time OS became more user-friendly and changed the way of people to interact with computer. GUI provides computer system visual elements which made user’s interaction with computer more comfortable and

user-friendly. User can just click on visual elements rather than typing commands. Here are some feature of GUI in Microsoft's windows icons, menus and windows.

6. Networked Systems – (1980s to 1990s)

At 1980s, the craze of computer networks at its peak. A special type of Operating Systems needed to manage the network communication. The OS like Novell NetWare and Windows NT were developed to manage network communication which provide users facility to work in collaborative environment and made file sharing and remote access very easy.

7. Mobile Operating Systems – (Late 1990s to Early 2000s)

Invention of smartphones create a big revolution in software industry, to handle the operation of smartphones, a special type of operating systems was developed. Some of them are: iOS and Android etc. These operating systems were optimized with the time and became more powerful.

8. AI Integration – (2010s to ongoing)

With the growth of time, Artificial intelligence came into picture. Operating system integrates features of AI technology like Siri, Google Assistant, and Alexa and became more powerful and efficient in many ways. These AI features with operating system create a entire new feature like voice commands, predictive text, and personalized recommendations.

Note: The above-mentioned OS basically tells how the OS evolved with the time by adding new features but it doesn't mean that only new generation OS are in use and previously OS system is not in use, according to the need, all these OS are still used in software industry.

Objective of OS

The primary objectives of an operating system are:

- To make computer system easier (virtual machine)
- Manages the resources of computer system

Virtual machine:

An operating system hides the detail of hardware resources from the programmer and the users and provides them with a convenient interface for using a computer system. It acts as an intermediary between hardware and its user, providing a high-level interface to low level hardware resources and making it easier for programmers and other users who use those resources.

Resource manager:

This objective of an operating system is to manage the various resources of the computer system. Thisinvolves keeping track of who is using resources, granting resource request, etc. the efficient and fair sharing of resources among the user and program is a key goal of operating system.

Function of OS

a) **Process management:**

The process management module of and operating system is responsible for the following task.

- Creation and deletion of user process
- Scheduling of system resources to different process requesting them.
- Providing mechanism for synchronization and communication among process
- Suspension and resumption of process

b) **Memory management:**

An OS has a Memory management module system which is responsible for the following task:

- Allocation and de-allocation of memory space to program.
- It also keeps the track of which memory segment is used by whom
- Manage for blank memory space available.
- Decides location for storage of new program.

c) **File management:**

File management module of and operating system is responsible for the following task.

- Creation and deletion of files and folders
- Backup files on non-volatile storage devices.
- Manage all files and folders and protect from unnecessary modification and deletion.
- Manage the naming convention of files and folders

d) **Security management:**

Security management module of and operating system is responsible for the following task.

- Protection and security of files and folders.
- Hiding system files from unwanted modification and deletion.
- Preventing theft, corruption and deletion of data.
- Ensuring authentication by prompting user name and password.

e) **Command interpretations:**

Command interpretation module of and operating system is responsible for the following task.

- To translate user commands into computer readable form
- Management of interface between computer user and hardware

f) Other related functions like:

- Act as a Resource Manager
- Device Management
- Input devices / Output devices
- Deadlock Prevention
- Time Management
- Coordinate with system software or hardware

Types of OS

Batch Processing Operating System

In a batch processing operating system interaction between the user and processor is limited or there is no interaction at all during the execution of work. Data and programs that need to be processed are bundled and collected as a 'batch' and executed together.

Batch processing operating systems are ideal in situations where:

- There are large amounts of data to be processed.
- Similar data needs to be processed.
- Similar processing is involved when executing the data.
- The system is capable of identifying times when the processor is idle at which time 'batches' maybe processed. Processing is all performed automatically without any user intervention.

Real-time Operating System

A real-time operating system processes inputs simultaneously, fast enough to affect the next input or process. Real-time systems are usually used to control complex systems that require a lot of processing like machinery and industrial systems. Real time operating systems are operating systems used in real time applications like embedded systems, robots, automobile engine controllers etc.

Example: Windows CE, RTL Linux, QNX, OS-9

Single User Operating System

A single user OS is designed for one user to effectively uses a computer at a time. Single user single tasking operating systems are those operating systems which can be used by a single user at a time. Also, we can only do a single task at a time in such operating systems.

Example: MS DOS. In Microsoft DOS only one program can be executed by a user at a time.

Multi-Tasking / Multi Programming Operating System

Multiprogramming operating system is an OS that is capable of loading and executing multiple programs simultaneously i.e. in such OS several applications may be simultaneously loaded and used in thememory. While the processor handles only one application at a particular time it is capable of switching between the applications effectively and simultaneously execute each application. This type of operating system can either be ***preemptive*** or ***cooperative***. In preemptive system the OS shares the CPU time and dedicates one slot to each of the programs. But in cooperative system is achieved by relying on each process to give time to the other processes in a defined manner but cooperative sharing is not used in recent modern OS.

Example: Windows operating system, UNIX

Time Sharing Operating System

Time sharing operating system is used for the effective utilization of resources such that it reduces the spare time of **CPU.** It allows the users to work on the same **CPU** at same time using time slot which provides equal time intervals for various process for their execution.

In **time sharing operating system**, equal time slots are provided to processor for execution of programs which also leads to the **context switching**, in which process shifts control from one to another. In case anyprocess does not complete its working within time slot then extra time slot will not be given to it i.e. it follows pre-emptive mechanism. For example- We are running a process with a time slot of 5 seconds andin case process does not complete in 5 second and requires 1 extra second then it will be executed in next execution cycle but time slot will not be extended.

Example: CTSS (Compatible Time-Sharing System)

Network Operating System

Network Operating System is the software that runs on a server and enables theserver to manage data, users, groups, security, applications, and other networking functions. It works under client server principle. The server provides services such as database and the client computer consume those services provided by the server.

The network operating system is designed to printer sharing, common file system and database sharing, application sharing, and the ability to manage a network name directory, security, and other housekeeping aspects of a network.

Example: Microsoft Windows Server 2003, Microsoft Windows Server 2008, UNIX, Linux, Mac OS X, Novell NetWare, and BSD.

Embedded Operating System:

It is designed to operate on some specific small devices such as PDA (Personal Digital Assistance), automatic electronic devices such as robot, TV Sets, Mobile Phones, etc and many more automatic digital equipment's. They are capable of operating with limited number of resources.

Examples: Minix3, Palm OS, Free BSD, etc.

Multi-User Operating System

This type of OS allows multiple users to simultaneously use the system, while here as well, the processor splits its resources and handles one user at a time, the speed and efficiency at which it does this makes it apparent that users are simultaneously using the system, some network systems utilize this kind of operating system.

Example: UNIX, Linux, IBM AS400

Distributed Operating System

In a distributed system, software and data may be distributed around the system, programs and files may be stored on different storage devices which are located in different geographical locations and may be accessed from different computer terminals.

A distributed operating system is a collection of autonomous computer systems capable of communicationand cooperation via their hardware and software interconnection. The distributed operating system provides an illusion to its users that it has a single uni-processor system, although it is actually consistedof multiple processors. A distributed operating system may facilitate access to remote resources, communication with remote processes, and distribution of computations

Example: UNIX, Windows 2000, Linux, etc

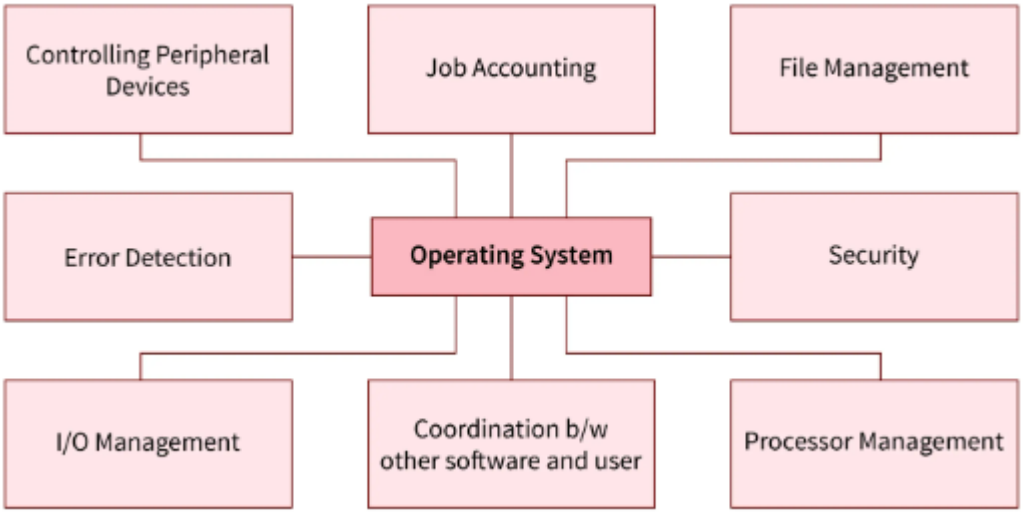
Architecture of Operating System

The operating system provides an environment for the users to execute computer programs. Operating systems are already installed on the computers you buy for e.g. personal computers have windows, Linux, and macOS, mainframe computers have z/OS, z/VM, etc., and mobile phones have operating systems such as Android, and iOS. The architecture of an operating system consists of four major components hardware, kernel, shell, and application and we shall explore all of them in detail one by one.

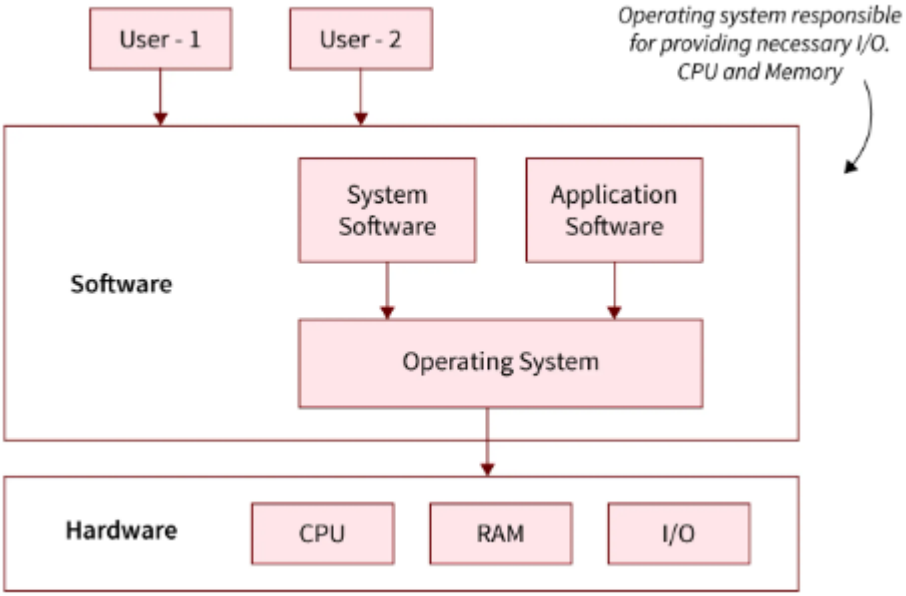
Highlights:

- The operating system gives an environment for executing programs by the users.
- Kernel is the most central part of the operating systems.
- Software running on any operating system can be system software and application

software. The operating system as we know is an intermediary and its functionalities include file management, memory management, process management, handling input and output, and peripheral devices as well.



The operating system handles all of the above tasks for the system and application software. The architecture of an operating system is basically the design of its software and hardware components. Depending upon the tasks or programs we need to run users can use the operating system most suitable for that program/software.



Before explaining various architectures of the operating systems, let's explore a few terms first which are part of the operating system.

1) Application: The application represents the software that a user is running on an operating system it can be either system or application software eg slack, sublime text editor, etc.

2) Shell: The shell represents software that provides an interface for the user where it serves to launch or start some program for which the user gives instructions.
It can be of two types first is a command line and another is a graphical user interface for eg: MS-DOS Shell, PowerShell, csh, ksh, etc.

3) Kernel: Kernel represents the most central and crucial part of the operating system where it is used for resource management i.e. it provides necessary I/O, processor, and memory to the application processes through inter-process communication mechanisms and system calls. Let's understand the various types of architectures of the operating system.

GUI and CUI Interface:

It is a platform where user can communicate with computer based on the user interface. Operating system canbe categorized into two:

- CUI or TUI (Text-based User Interface)
- GUI (Graphical User Interface)

CUI or TUI (Text-based User Interface)

CUI stands for command user interface and it is the earlier form of operating system.In such operating system primary method of communication with the computer is command or line of text. The user has to learn large set of commands to run the program. The users can't view any graphics painting devices like mouse and icons facilities are not available. It is extremely difficult and boring for the general users to work on the CUI based operating system.
Example: MS-DOS

GUI (Graphical User Interface)

A visual computing environment that represents program files and options with graphical image such as icons, menus and dialog boxes on the screen is called GUI. A small image displayed on the screen to present on object that can be manipulated by the user is called icon. User can select or activate these options by pointing and clicking with the mouse or keyword. GUI provides a framework to use scroll bar, painting arrow for mouse, select criteria by highlighting on it to handle user actions. Different list box, check box, buttons, windows facilities are available in GUI based operating system, and due to these various facilities, it is popular among computer users than that of CUI based operating system. Example: Windows

There are differences between the CUI and the GUI are as follows:

Features	CUI	GUI
Full-Form	CUI stands for Character User Interface.	GUI stands for Graphical User Interface.
Interaction	The user interacts with the computer using commands like text.	The user interacts with the system using Graphics like icons, images.
Navigation	Navigation is not easy.	Navigation is easy to use.
Usage	Usage is easy to use.	Usage is difficult, requires expertise.
Speed	It has high speed.	It has a low speed.
Memory Requirement	It has a low memory requirement.	It has a high memory requirement.
Peripherals used	Users interact with the computer system by typing commands into the keyboard.	Users interact with the computer system using a graphical interface, which includes menus and mouse clicks.
Precision	It has high precision.	It has low precision.
Flexibility	It has a little flexible user interface.	It has a highly flexible user interface.
Customize	It is not easily changeable.	It has highly customizable.

The booting system Windows/ Linux operating system, settings, properties and installation guides

You have to prepare notes on this topic based on the task you did in the workshop laboratories.